

Ex. 2.27

$$X: \mu=25, \sigma^2=16.$$

$$\begin{aligned} \Rightarrow P(19 \leq X \leq 33) &= P(|X-25| \leq 8) \\ &= P(|X-\mu| \leq 2\sigma) \leadsto \underline{k=2}. \\ &\geq 0.95. \end{aligned}$$

Thm. 2.16 (Cauchy-Schwarz)

$$\begin{aligned} \text{If } E[X^2] < \infty, E[Y^2] < \infty, \\ \Rightarrow (E[XY])^2 &\leq E[X^2]E[Y^2]. \end{aligned}$$

"=" holds when $P(Y=cX)=1$ for some c .

(pf). Let $h(t) := E[(tX-Y)^2] = E[X^2]t^2 - 2E[XY]t + E[Y^2]$.

Then $h(t) \geq 0$ (quadratic fn.).

If $h(t) > 0$, then $D = 4(E[XY])^2 - 4E[X^2]E[Y^2] < 0$.

If $h(t) = E[(tX-Y)^2] = 0$ holds when $t=t_0$,

then $E[(t_0X-Y)^2] = 0$ and $(t_0X-Y)^2 \geq 0 \leadsto P(t_0X-Y)=1$. $\square$

2.5 Probability distribution of discrete r.v.

$$X \sim \text{Bernoulli}(p) \Rightarrow \text{pmf: } f(x) = \begin{cases} p, & x=1 \\ 1-p, & x=0. \end{cases}$$

$$\Rightarrow E[X] = 1 \times p + 0 \times (1-p) = p,$$

$$\begin{aligned} \text{Var}(X) &= E[(X-\mu)^2] = E[(X-p)^2] = (0-p)^2 \times (1-p) + (1-p)^2 \times p \\ &= p(1-p). \end{aligned}$$

(2).

Indicator ftn.

For $s \in S$, $X(s) = I_A(s) \Rightarrow X \sim \text{Bernoulli}(p)$
s.t. $p = P(A) = P(X=1)$.

$$\text{i.e., } I_A(s) = \begin{cases} 1, & s \in A \\ 0, & s \in A^c \end{cases} \quad (\text{sometimes } "I(s \in A)").$$

Binomial Distribution.

$X_1, \dots, X_n \sim \text{Bernoulli}(p)$ & independent.

$\Rightarrow X = \sum_{i=1}^n X_i$: "binomial random variable."

$$\Rightarrow \text{pmf: } f(x) = \binom{n}{x} p^x (1-p)^{n-x}. \quad (x=0, \dots, n).$$

Denote as $X \sim B(n, p)$.

$$\Rightarrow E[X] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = np.$$

$$\Rightarrow \text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = np(1-p).$$

Multinomial Dist.

k -mutually exclusive events A_i . s.t. $\bigcup_{i=1}^k A_i = S$. and $P(A_i) = p_i (i=1, \dots, k)$

#(A_i happens) be X_i . Then the combined prob. dist. ftn is:
(n times trial, indep.)

$$f(x_1, x_2, \dots, x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

$$\left(\sum_{i=1}^k p_i = 1, \quad x_i: 0 \leq x_i \leq n \text{ \& } \sum_{i=1}^k x_i = n \right).$$

(3)

If the obs. of $\mathbf{X} = (X_1, X_2, \dots, X_k)$ is $(x_1, x_2, \dots, x_k)$, then

A_1 observed x_1 times,

A_2 " x_2 " ,

$\vdots$

A_k " x_k " ~ on n times trials.

$\Leftrightarrow$ divide k -group of size $x_1, \dots, x_k$.

$$\Rightarrow \binom{n}{x_1, x_2, \dots, x_k} = \frac{n!}{x_1! x_2! \dots x_k!}$$

Certain column happens with the prob. of $p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$.

$$\Rightarrow P((X_1, \dots, X_k) = (x_1, \dots, x_k)) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}$$

$\Rightarrow$ the random vector $\mathbf{X} = (X_1, \dots, X_k)$ follows

"multinomial distribution". Denote: $\mathbf{X} \sim \text{Mult}(n, p_1, \dots, p_k)$.

Ex. 2.31 Let $(X_1, X_2, X_3) \sim \text{Mult}(n, p_1, p_2, p_3)$.

and $X_1 \sim B(n, p_1)$, $X_2 \sim B(n, p_2)$, $X_2 | X_1 = x_1 \sim B(n - x_1, p)$,

$$p = \frac{p_2}{1 - p_1} \quad \text{By Thm 2.11,}$$

$$E[X_1 X_2] = E[E[X_1 X_2 | X_1]]$$

$$= E[X_1 E[X_2 | X_1]] = E\left[\frac{X_1(n - X_1)p_2}{1 - p_1}\right]$$

$$= \left[\frac{p_2}{1 - p_1}\right] [nE[X_1] - E[X_1^2]]$$

$$= n(n-1)p_1 p_2.$$

④

$$\begin{aligned}\Rightarrow \text{Cov}(X_1, X_2) &= E[X_1 X_2] - E[X_1]E[X_2] \\ &= n(n-1)p_1 p_2 - (np_1)(np_2) \\ &= -np_1 p_2 \dots\end{aligned}$$

(2.5.3) Poisson dist.

3 assumptions.

- $$\left\{ \begin{array}{l} (1) \lim_{h \rightarrow 0} P(\text{일회 발생 } h \text{ 시간 내에 사건 } A \text{가 1회 일어남})/h = \lambda. \\ (2) \lim_{h \rightarrow 0} P(\text{ " " " " 2회 이상 일어남})/h = 0. \\ (3) \text{ 서로 겹치지 않는 두 구간 내에서 사건 } A \text{의 발생 횟수는 독립.} \end{array} \right.$$

$\vdots$

$$P(X(t)=x) \sim f(x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x=0, 1, 2, \dots$$

If $t=1$,

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x=0, 1, 2, \dots \Rightarrow X \sim \text{Poi}(\lambda).$$

"Poisson Process."

unit time $\rightarrow \text{Poi}(\lambda)$.

$(t_1, t_2] \rightarrow \text{Poi}(\lambda(t_2 - t_1))$.

$$\Rightarrow E[X] = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda.$$

$$\begin{aligned}\text{Var}(X) \Rightarrow E[X(X-1)] &= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} = \lambda^2 e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = \lambda^2.\end{aligned}$$

$$\begin{aligned}\Rightarrow E[X^2] &= \lambda^2 + \lambda. \Rightarrow \text{Var}(X) = E[X^2] - (E[X])^2 \\ &= (\lambda^2 + \lambda) - (\lambda)^2 = \lambda.\end{aligned}$$

(5)

Thm. 2.17 (Binomial dist. approximation by Poisson dist.)

Let $X \sim B(n, p)$. Let $p \rightarrow 0$ as $n \rightarrow \infty$, s.t. $np = \lambda$.

for $x = 0, 1, 2, \dots$

$$\lim_{n \rightarrow \infty} \binom{n}{x} p^x (1-p)^{n-x} = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{holds.}$$

$$\begin{aligned} \text{(pf). } \lim_{n \rightarrow \infty} \binom{n}{x} p^x (1-p)^{n-x} &= \lim_{n \rightarrow \infty} \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \lim_{n \rightarrow \infty} \frac{n!}{x!(n-x)!} \lambda^x \left(\frac{1}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \left(1 - \frac{\lambda}{n}\right)^x \\ &= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \frac{n!}{(n-x)! (n-x)^x} \left(1 - \frac{\lambda}{n}\right)^n \\ &\Rightarrow \frac{n!}{(n-x)! (n-x)^x} = \frac{n!}{(n-x)! n^x \left(1 - \frac{\lambda}{n}\right)^x} \\ &= \frac{n(n-1) \dots (n-x+1)}{n^x \left(1 - \frac{\lambda}{n}\right)^x} \\ &= 1 \times \left(1 - \frac{\lambda}{n}\right) \times \left(1 - \frac{\lambda}{n}\right) \times \dots \times \left(1 - \frac{\lambda}{n}\right) \times \left(1 - \frac{\lambda}{n}\right)^{-x} \\ &\Rightarrow \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda} \end{aligned}$$

□

2.5.4 Geometric dist.

Success prob: p , first success. $\Rightarrow X$.

$$\text{pdf: } f(x) = p(1-p)^{x-1} \quad (x=1, 2, \dots)$$

" $X \sim \text{Geo}(p)$ "

Q. $E[X] = \sum_{x=1}^{\infty} x p (1-p)^{x-1}$ (Let $q := 1-p$.)

$$\begin{aligned}
 &= \sum_{x=1}^{\infty} p \frac{d}{dq} q^x \\
 &= p \frac{d}{dq} \sum_{x=1}^{\infty} q^x \\
 &= p \frac{d}{dq} \frac{q}{1-q} \\
 &= p \cdot (1-q)^{-2} = \frac{1}{p} \quad \text{..}
 \end{aligned}$$

$$\begin{aligned}
 E[X(X+1)] &= \sum_{x=1}^{\infty} x(x+1) p q^{x-1} \\
 &= p \sum_{x=1}^{\infty} \frac{d^2}{dq^2} q^{x+1} \\
 &= p \frac{d^2}{dq^2} \sum_{x=1}^{\infty} q^{x+1} \\
 &= p \frac{d^2}{dq^2} \{ (1-q)^{-1} q^2 \} = \frac{2}{p^2}
 \end{aligned}$$

$$\Rightarrow E[X^2] = \frac{2}{p^2} - \frac{1}{p} = \frac{1+q}{p^2}$$

$$\Rightarrow \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1+q}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2} \quad \text{..}$$

Thm. 2.18 (Memoryless Property)

If $X \sim \text{Geo}(p)$, then $\forall j, k \in \mathbb{N}$, $P(X > j+k | X > j) = P(X > k)$.

(pf). $P(X > k) = \sum_{j=k+1}^{\infty} p(1-p)^{j-1} = (1-p)^k \sum_{j=1}^{\infty} p(1-p)^{j-1} = (1-p)^k$.

$$\Rightarrow P(X > j+k | X > j) = \frac{P(X > j+k)}{P(X > j)} = \frac{(1-p)^{j+k}}{(1-p)^j} = (1-p)^k = P(X > k). \quad \square$$

Note. $P(X > j+k) = P(X > j) \cdot P(X > k)$.